betaMC: Internal Tests

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Tests

```
#> test-betaMC-beta-mc-est-mi
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0740 5 0.4391 0.4393 0.4400 0.5950 0.5963 0.5966
#> PCTGRT  0.3915 0.0713 5 0.2236 0.2240 0.2254 0.3917 0.4014 0.4035
#> PCTSUPP 0.2632 0.0815 5 0.1581 0.1607 0.1726 0.3626 0.3653 0.3659
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0740 5 0.4391 0.4393 0.4400 0.5950 0.5963 0.5966
#> PCTGRT  0.3915 0.0713 5 0.2236 0.2240 0.2254 0.3917 0.4014 0.4035
#> PCTSUPP 0.2632 0.0815 5 0.1581 0.1607 0.1726 0.3626 0.3653 0.3659
#> Test passed with 1 success.
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>           est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>           est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
```

```
#> Test passed with 1 success.
```

#> test-betaMC-beta-mc-est

```
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0740 5 0.4391 0.4393 0.4400 0.5950 0.5963 0.5966
#> PCTGRT  0.3915 0.0713 5 0.2236 0.2240 0.2254 0.3917 0.4014 0.4035
#> PCTSUPP 0.2632 0.0815 5 0.1581 0.1607 0.1726 0.3626 0.3653 0.3659
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4951 0.0740 5 0.4391 0.4393 0.4400 0.5950 0.5963 0.5966
#> PCTGRT  0.3915 0.0713 5 0.2236 0.2240 0.2254 0.3917 0.4014 0.4035
#> PCTSUPP 0.2632 0.0815 5 0.1581 0.1607 0.1726 0.3626 0.3653 0.3659
#> Test passed with 1 success.
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>           est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Call:
#> BetaMC(object = mc)
#>
#> Standardized regression slopes
#> type = "mvn"
#>           est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC 0.7622 0.0793 5 0.6755 0.6766 0.6815 0.8774 0.8837 0.8851
#> Test passed with 1 success.
```

#> test-betaMC-delta-r-sq-mc-est-mi

```
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
```

```
#> NARTIC  0.1859 0.0674 5 0.0976 0.0982 0.1009 0.2662 0.2731 0.2746
#> PCTGRT  0.1177 0.0309 5 0.0389 0.0389 0.0391 0.1065 0.1107 0.1117
#> PCTSUPP 0.0569 0.0300 5 0.0146 0.0152 0.0178 0.0875 0.0882 0.0884
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.1859 0.0674 5 0.0976 0.0982 0.1009 0.2662 0.2731 0.2746
#> PCTGRT  0.1177 0.0309 5 0.0389 0.0389 0.0391 0.1065 0.1107 0.1117
#> PCTSUPP 0.0569 0.0300 5 0.0146 0.0152 0.0178 0.0875 0.0882 0.0884
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-delta-r-sq-mc-est

```
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.1859 0.0674 5 0.0976 0.0982 0.1009 0.2662 0.2731 0.2746
#> PCTGRT  0.1177 0.0309 5 0.0389 0.0389 0.0391 0.1065 0.1107 0.1117
#> PCTSUPP 0.0569 0.0300 5 0.0146 0.0152 0.0178 0.0875 0.0882 0.0884
#> Call:
#> DeltaRSqMC(object = mc)
#>
#> Improvement in R-squared
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.1859 0.0674 5 0.0976 0.0982 0.1009 0.2662 0.2731 0.2746
#> PCTGRT  0.1177 0.0309 5 0.0389 0.0389 0.0391 0.1065 0.1107 0.1117
#> PCTSUPP 0.0569 0.0300 5 0.0146 0.0152 0.0178 0.0875 0.0882 0.0884
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-diff-beta-mc-est-mi

```
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1273 5  0.0356  0.0384  0.0505 0.3655 0.3715 0.3729
```

```
#> NARTIC-PCTSUPP 0.2319 0.1426 5  0.0819  0.0824  0.0846 0.4088 0.4194 0.4217
#> PCTGRT-PCTSUPP 0.1282 0.1070 5 -0.1052 -0.1050 -0.1038 0.1197 0.1238 0.1247
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1273 5  0.0356  0.0384  0.0505 0.3655 0.3715 0.3729
#> NARTIC-PCTSUPP 0.2319 0.1426 5  0.0819  0.0824  0.0846 0.4088 0.4194 0.4217
#> PCTGRT-PCTSUPP 0.1282 0.1070 5 -0.1052 -0.1050 -0.1038 0.1197 0.1238 0.1247
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-diff-beta-mc-est

#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1273 5  0.0356  0.0384  0.0505 0.3655 0.3715 0.3729
#> NARTIC-PCTSUPP 0.2319 0.1426 5  0.0819  0.0824  0.0846 0.4088 0.4194 0.4217
#> PCTGRT-PCTSUPP 0.1282 0.1070 5 -0.1052 -0.1050 -0.1038 0.1197 0.1238 0.1247
#> Call:
#> DiffBetaMC(object = mc)
#>
#> Differences of standardized regression slopes
#> type = "mvn"
#>                   est     se R   0.05%    0.5%    2.5%  97.5%  99.5% 99.95%
#> NARTIC-PCTGRT  0.1037 0.1273 5  0.0356  0.0384  0.0505 0.3655 0.3715 0.3729
#> NARTIC-PCTSUPP 0.2319 0.1426 5  0.0819  0.0824  0.0846 0.4088 0.4194 0.4217
#> PCTGRT-PCTSUPP 0.1282 0.1070 5 -0.1052 -0.1050 -0.1038 0.1197 0.1238 0.1247
#> Test passed with 1 success.
#> Test passed with 1 success.

#> test-betaMC-mc-fixed-x-mi

#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
```

```
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-mc-fixed-x

```
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> MC(object = object, R = 5L, decomposition = "chol", fixed_x = TRUE)
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4758462 0.4987145
#>
#> $sigmasq
#> [1] 0.5337949
#>
#> $vechsigmacapx
#> [1] 1.000000e+00 6.101989e-17 1.000000e+00
#>
#> $sigmacapx
#>              [,1]         [,2]
#> [1,] 1.000000e+00 6.101989e-17
#> [2,] 6.101989e-17 1.000000e+00
#>
#> $sigmaysq
#> [1] 1.008941
#>
#> $sigmayx
#> [1] 0.4758462 0.4987145
#>
#> $sigmacap
#>           [,1]         [,2]         [,3]
#> [1,] 1.0089407 4.758462e-01 4.987145e-01
#> [2,] 0.4758462 1.000000e+00 6.101989e-17
#> [3,] 0.4987145 6.101989e-17 1.000000e+00
#>
#> $pd
#> [1] TRUE
```

```
#>
#> Call:
#> MC(object = object, R = 5L, decomposition = "svd", fixed_x = TRUE)
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4659576 0.5142679
#>
#> $sigmasq
#> [1] 0.5359596
#>
#> $vechsigmacapx
#> [1] 1.000000e+00 6.101989e-17 1.000000e+00
#>
#> $sigmacapx
#>              [,1]         [,2]
#> [1,] 1.000000e+00 6.101989e-17
#> [2,] 6.101989e-17 1.000000e+00
#>
#> $sigmaysq
#> [1] 1.017548
#>
#> $sigmayx
#> [1] 0.4659576 0.5142679
#>
#> $sigmacap
#>           [,1]         [,2]         [,3]
#> [1,] 1.0175476 4.659576e-01 5.142679e-01
#> [2,] 0.4659576 1.000000e+00 6.101989e-17
#> [3,] 0.5142679 6.101989e-17 1.000000e+00
#>
#> $pd
#> [1] TRUE
#>
#> Test passed with 1 success.
```

#> test-betaMC-mc-mi

```
#> MCMI(object = object, mi = mi, R = R, type = "mvn")
#> $mean
#>       b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>   0.4826    0.4891    0.5314    1.0038    0.0097    0.9983
#>
#> $var
#>              b1     b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1        5e-04  0e+00   0e+00    0.0000     0e+00     1e-04
```

```
#> b2        0e+00  7e-04  -1e-04   -0.0002    -1e-04     1e-04
#> sigmasq   0e+00 -1e-04   4e-04    0.0000     0e+00     1e-04
#> sigmax1x1 0e+00 -2e-04   0e+00    0.0019    -2e-04     3e-04
#> sigmax2x1 0e+00 -1e-04   0e+00   -0.0002     1e-03    -1e-04
#> sigmax2x2 1e-04  1e-04   1e-04    0.0003    -1e-04     2e-03
#>
#> $bias
#>       b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>  -0.0020    0.0019    0.0008    0.0038   -0.0003   -0.0017
#>
#> $rmse
#>       b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>   0.0220    0.0257    0.0207    0.0430    0.0318    0.0441
#>
#> $location
#>       b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>   0.4845    0.4872    0.5307    1.0000    0.0100    1.0000
#>
#> $scale
#>             b1    b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1        5e-04 0e+00   0e+00     0.000     0.000     0.000
#> b2        0e+00 5e-04   0e+00     0.000     0.000     0.000
#> sigmasq   0e+00 0e+00   6e-04     0.000     0.000     0.000
#> sigmax1x1 0e+00 0e+00   0e+00     0.002     0.000     0.000
#> sigmax2x1 0e+00 0e+00   0e+00     0.000     0.001     0.000
#> sigmax2x2 0e+00 0e+00   0e+00     0.000     0.000     0.002
#>
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-mc

```
#> MC(object = object, R = R, type = "mvn")
#> $mean
#>       b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>   0.4838    0.4895    0.5317    1.0068    0.0126    0.9946
#>
#> $var
#>              b1     b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1         6e-04  1e-04  -1e-04    0.0000   -0.0001    0.0000
```

```
#> b2         1e-04  5e-04   1e-04    0.0001    0.0000   -0.0002
#> sigmasq   -1e-04  1e-04   6e-04    0.0001   -0.0001    0.0001
#> sigmax1x1  0e+00  1e-04   1e-04    0.0018   -0.0002   -0.0002
#> sigmax2x1 -1e-04  0e+00  -1e-04   -0.0002    0.0012   -0.0002
#> sigmax2x2  0e+00 -2e-04   1e-04   -0.0002   -0.0002    0.0018
#>
#> $bias
#>        b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>   -0.0008    0.0023    0.0010    0.0068    0.0026   -0.0054
#>
#> $rmse
#>        b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>    0.0236    0.0224    0.0248    0.0433    0.0342    0.0425
#>
#> $location
#>        b1        b2   sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#>    0.4845    0.4872    0.5307    1.0000    0.0100    1.0000
#>
#> $scale
#>              b1    b2 sigmasq sigmax1x1 sigmax2x1 sigmax2x2
#> b1        5e-04 0e+00   0e+00     0.000     0.000     0.000
#> b2        0e+00 5e-04   0e+00     0.000     0.000     0.000
#> sigmasq   0e+00 0e+00   6e-04     0.000     0.000     0.000
#> sigmax1x1 0e+00 0e+00   0e+00     0.002     0.000     0.000
#> sigmax2x1 0e+00 0e+00   0e+00     0.000     0.001     0.000
#> sigmax2x2 0e+00 0e+00   0e+00     0.000     0.000     0.002
#>
#> Test passed with 9 successes.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Test passed with 1 success.
#> Call:
#> MC(object = object, R = 5L, decomposition = "chol")
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4688013 0.4660633
#>
#> $sigmasq
#> [1] 0.5331191
```

```
#>
#> $vechsigmacapx
#> [1]  0.9497147989 -0.0003547646  1.0075414443
#>
#> $sigmacapx
#>               [,1]          [,2]
#> [1,]  0.9497147989 -0.0003547646
#> [2,] -0.0003547646  1.0075414443
#>
#> $sigmaysq
#> [1] 0.9605405
#>
#> $sigmayx
#> [1] 0.4450622 0.4694118
#>
#> $sigmacap
#>           [,1]          [,2]          [,3]
#> [1,] 0.9605405  0.4450622080  0.4694118229
#> [2,] 0.4450622  0.9497147989 -0.0003547646
#> [3,] 0.4694118 -0.0003547646  1.0075414443
#>
#> $pd
#> [1] TRUE
#>
#> Call:
#> MC(object = object, R = 5L, decomposition = "svd")
#> The first set of simulated parameter estimates
#> and model-implied covariance matrix.
#>
#> $coef
#> [1] 0.4580301 0.4850555
#>
#> $sigmasq
#> [1] 0.5497989
#>
#> $vechsigmacapx
#> [1] 1.038296916 0.009804769 1.068440723
#>
#> $sigmacapx
#>             [,1]        [,2]
#> [1,] 1.038296916 0.009804769
#> [2,] 0.009804769 1.068440723
#>
#> $sigmaysq
#> [1] 1.023363
#>
```

```
#> $sigmayx
#> [1] 0.480327 0.522744
#>
#> $sigmacap
#>          [,1]        [,2]        [,3]
#> [1,] 1.023363 0.480327047 0.522743966
#> [2,] 0.480327 1.038296916 0.009804769
#> [3,] 0.522744 0.009804769 1.068440723
#>
#> $pd
#> [1] TRUE
#>
#> Test passed with 1 success.
```

#> test-betaMC-p-cor-mc-est-mi

```
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1011 5 0.3254 0.3263 0.3307 0.5745 0.5823 0.5841
#> PCTGRT  0.3757 0.0968 5 0.1659 0.1660 0.1662 0.3747 0.3896 0.3930
#> PCTSUPP 0.2254 0.0943 5 0.0685 0.0714 0.0841 0.3115 0.3149 0.3157
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1011 5 0.3254 0.3263 0.3307 0.5745 0.5823 0.5841
#> PCTGRT  0.3757 0.0968 5 0.1659 0.1660 0.1662 0.3747 0.3896 0.3930
#> PCTSUPP 0.2254 0.0943 5 0.0685 0.0714 0.0841 0.3115 0.3149 0.3157
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-p-cor-mc-est

```
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1011 5 0.3254 0.3263 0.3307 0.5745 0.5823 0.5841
#> PCTGRT  0.3757 0.0968 5 0.1659 0.1660 0.1662 0.3747 0.3896 0.3930
```

```
#> PCTSUPP 0.2254 0.0943 5 0.0685 0.0714 0.0841 0.3115 0.3149 0.3157
#> Call:
#> PCorMC(object = mc)
#>
#> Squared partial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4874 0.1011 5 0.3254 0.3263 0.3307 0.5745 0.5823 0.5841
#> PCTGRT  0.3757 0.0968 5 0.1659 0.1660 0.1662 0.3747 0.3896 0.3930
#> PCTSUPP 0.2254 0.0943 5 0.0685 0.0714 0.0841 0.3115 0.3149 0.3157
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-r-sq-mc-est-mi

```
#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0435 5 0.7285 0.7298 0.7357 0.8393 0.8404 0.8406
#> adj 0.7906 0.0466 5 0.7091 0.7105 0.7168 0.8279 0.8290 0.8292
#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0435 5 0.7285 0.7298 0.7357 0.8393 0.8404 0.8406
#> adj 0.7906 0.0466 5 0.7091 0.7105 0.7168 0.8279 0.8290 0.8292
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-r-sq-mc-est

```
#> Call:
#> RSqMC(object = mc)
#>
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0435 5 0.7285 0.7298 0.7357 0.8393 0.8404 0.8406
#> adj 0.7906 0.0466 5 0.7091 0.7105 0.7168 0.8279 0.8290 0.8292
#> Call:
#> RSqMC(object = mc)
#>
```

```
#> R-squared and adjusted R-squared
#> type = "mvn"
#>        est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> rsq 0.8045 0.0435 5 0.7285 0.7298 0.7357 0.8393 0.8404 0.8406
#> adj 0.7906 0.0466 5 0.7091 0.7105 0.7168 0.8279 0.8290 0.8292
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-s-cor-mc-est-mi

```
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0799 5 0.3124 0.3133 0.3173 0.5152 0.5224 0.5240
#> PCTGRT  0.3430 0.0579 5 0.1972 0.1973 0.1976 0.3251 0.3325 0.3341
#> PCTSUPP 0.2385 0.0720 5 0.1209 0.1226 0.1303 0.2958 0.2970 0.2973
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0799 5 0.3124 0.3133 0.3173 0.5152 0.5224 0.5240
#> PCTGRT  0.3430 0.0579 5 0.1972 0.1973 0.1976 0.3251 0.3325 0.3341
#> PCTSUPP 0.2385 0.0720 5 0.1209 0.1226 0.1303 0.2958 0.2970 0.2973
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC-s-cor-mc-est

```
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0799 5 0.3124 0.3133 0.3173 0.5152 0.5224 0.5240
#> PCTGRT  0.3430 0.0579 5 0.1972 0.1973 0.1976 0.3251 0.3325 0.3341
#> PCTSUPP 0.2385 0.0720 5 0.1209 0.1226 0.1303 0.2958 0.2970 0.2973
#> Call:
#> SCorMC(object = mc)
#>
#> Semipartial correlations
#> type = "mvn"
```

```
#>            est     se R  0.05%   0.5%   2.5%  97.5%  99.5% 99.95%
#> NARTIC  0.4312 0.0799 5 0.3124 0.3133 0.3173 0.5152 0.5224 0.5240
#> PCTGRT  0.3430 0.0579 5 0.1972 0.1973 0.1976 0.3251 0.3325 0.3341
#> PCTSUPP 0.2385 0.0720 5 0.1209 0.1226 0.1303 0.2958 0.2970 0.2973
#> Test passed with 1 success.
#> Test passed with 1 success.
```

#> test-betaMC

```
#> Test passed with 1 success.
```

#> test-zzz-coverage

```
#>               beta1    beta2    beta3 sigmasq   sigmax1x1  sigmax2x1  sigmax3x1
#> sigmaysq   909.1981 257.2976 276.0367       1 0.007091036 0.03637752 0.01896371
#> sigmayx1  3507.1691 471.2058 510.5430       0 0.084208291 0.21599726 0.11260003
#> sigmayx2   471.2058 333.2295 150.9121       0 0.000000000 0.08420829 0.00000000
#> sigmayx3   510.5430 150.9121 554.4386       0 0.000000000 0.00000000 0.08420829
#> sigmax1x1    0.0000   0.0000   0.0000       0 1.000000000 0.00000000 0.00000000
#> sigmax2x1    0.0000   0.0000   0.0000       0 0.000000000 1.00000000 0.00000000
#> sigmax3x1    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 1.00000000
#> sigmax2x2    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#> sigmax3x2    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#> sigmax3x3    0.0000   0.0000   0.0000       0 0.000000000 0.00000000 0.00000000
#>            sigmax2x2 sigmax3x2  sigmax3x3
#> sigmaysq  0.04665482 0.0486426 0.01267877
#> sigmayx1  0.00000000 0.0000000 0.00000000
#> sigmayx2  0.21599726 0.1126000 0.00000000
#> sigmayx3  0.00000000 0.2159973 0.11260003
#> sigmax1x1 0.00000000 0.0000000 0.00000000
#> sigmax2x1 0.00000000 0.0000000 0.00000000
#> sigmax3x1 0.00000000 0.0000000 0.00000000
#> sigmax2x2 1.00000000 0.0000000 0.00000000
#> sigmax3x2 0.00000000 1.0000000 0.00000000
#> sigmax3x3 0.00000000 0.0000000 1.00000000
#>               beta1    beta2    beta3 sigmasq
#> sigmaysq   909.1981 257.2976 276.0367       1
#> sigmayx1  3507.1691 471.2058 510.5430       0
#> sigmayx2   471.2058 333.2295 150.9121       0
#> sigmayx3   510.5430 150.9121 554.4386       0
#> sigmax1x1    0.0000   0.0000   0.0000       0
#> sigmax2x1    0.0000   0.0000   0.0000       0
#> sigmax3x1    0.0000   0.0000   0.0000       0
#> sigmax2x2    0.0000   0.0000   0.0000       0
#> sigmax3x2    0.0000   0.0000   0.0000       0
#> sigmax3x3    0.0000   0.0000   0.0000       0
#>               beta1    beta2    beta3       rsq   sigmax1x1  sigmax2x1
```

```
#> sigmaysq   909.1981 257.2976 276.0367 -126.0843 0.007091036 0.03637752
#> sigmayx1  3507.1691 471.2058 510.5430    0.0000 0.084208291 0.21599726
#> sigmayx2   471.2058 333.2295 150.9121    0.0000 0.000000000 0.08420829
#> sigmayx3   510.5430 150.9121 554.4386    0.0000 0.000000000 0.00000000
#> sigmax1x1    0.0000   0.0000   0.0000    0.0000 1.000000000 0.00000000
#> sigmax2x1    0.0000   0.0000   0.0000    0.0000 0.000000000 1.00000000
#> sigmax3x1    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax2x2    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax3x2    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#> sigmax3x3    0.0000   0.0000   0.0000    0.0000 0.000000000 0.00000000
#>            sigmax3x1  sigmax2x2 sigmax3x2  sigmax3x3
#> sigmaysq  0.01896371 0.04665482 0.0486426 0.01267877
#> sigmayx1  0.11260003 0.00000000 0.0000000 0.00000000
#> sigmayx2  0.00000000 0.21599726 0.1126000 0.00000000
#> sigmayx3  0.08420829 0.00000000 0.2159973 0.11260003
#> sigmax1x1 0.00000000 0.00000000 0.0000000 0.00000000
#> sigmax2x1 0.00000000 0.00000000 0.0000000 0.00000000
#> sigmax3x1 1.00000000 0.00000000 0.0000000 0.00000000
#> sigmax2x2 0.00000000 1.00000000 0.0000000 0.00000000
#> sigmax3x2 0.00000000 0.00000000 1.0000000 0.00000000
#> sigmax3x3 0.00000000 0.00000000 0.0000000 1.00000000
#>               beta1    beta2    beta3       rsq
#> sigmaysq   909.1981 257.2976 276.0367 -126.0843
#> sigmayx1  3507.1691 471.2058 510.5430    0.0000
#> sigmayx2   471.2058 333.2295 150.9121    0.0000
#> sigmayx3   510.5430 150.9121 554.4386    0.0000
#> sigmax1x1    0.0000   0.0000   0.0000    0.0000
#> sigmax2x1    0.0000   0.0000   0.0000    0.0000
#> sigmax3x1    0.0000   0.0000   0.0000    0.0000
#> sigmax2x2    0.0000   0.0000   0.0000    0.0000
#> sigmax3x2    0.0000   0.0000   0.0000    0.0000
#> sigmax3x3    0.0000   0.0000   0.0000    0.0000
#> Test passed with 1 success.
#> [[1]]
#> [[1]][[1]]
#> [[1]][[1]]$value
#> [[1]][[1]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[1]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[2]]
#> [[1]][[2]]$value
```

```
#> [[1]][[2]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[2]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[3]]
#> [[1]][[3]]$value
#> [[1]][[3]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[3]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[4]]
#> [[1]][[4]]$value
#> [[1]][[4]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[4]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[5]]
#> [[1]][[5]]$value
#> [[1]][[5]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[5]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[6]]
#> [[1]][[6]]$value
#> [[1]][[6]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[6]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[7]]
#> [[1]][[7]]$value
#> [[1]][[7]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[7]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[8]]
#> [[1]][[8]]$value
#> [[1]][[8]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[8]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[9]]
#> [[1]][[9]]$value
#> [[1]][[9]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[9]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[10]]
#> [[1]][[10]]$value
#> [[1]][[10]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[10]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[11]]
#> [[1]][[11]]$value
```

```
#> [[1]][[11]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[11]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[12]]
#> [[1]][[12]]$value
#> [[1]][[12]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[12]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[13]]
#> [[1]][[13]]$value
#> [[1]][[13]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[13]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[14]]
#> [[1]][[14]]$value
#> [[1]][[14]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[14]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[15]]
#> [[1]][[15]]$value
#> [[1]][[15]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[15]]$visible
```

```
#> [1] TRUE
#>
#>
#> [[1]][[16]]
#> [[1]][[16]]$value
#> [[1]][[16]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[16]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[17]]
#> [[1]][[17]]$value
#> [[1]][[17]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[17]]$visible
#> [1] TRUE
#>
#>
#> [[1]][[18]]
#> [[1]][[18]]$value
#> [[1]][[18]]$value[[1]]
#> [1] TRUE
#>
#>
#> [[1]][[18]]$visible
#> [1] TRUE
```

Environment

```
ls()
```

```
#> [1] "nas1982" "root"
```

Class

```
#> [[1]]
#> [1] "data.frame"
#>
#> [[2]]
#> [1] "root_criterion"
```

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/